A Slater-adapted oscillator in complex double-scaled SYK

Same-disorder polarization, all fixed vacuum words, and the first Hamiltonian defect

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Abstract

We select the pure particle–hole creation block of the original complex SYK Hamiltonian relative to a half-filled Slater state. All fixed ordered vacuum moments of its normalized creation and annihilation operators converge to a single dressed bosonic oscillator, $M = q^{\hat{n}/2}b$, $[b, b^\dagger] = 1$, with $q = e^{-4\lambda}$. The proof counts all forward Gaussian pairings and bounds the finite-size backward pairings rather than discarding them. The construction implements the state-adapted strategy of recent pure-state DSSYK work in a charge-conserving, same-disorder model. Microscopic occupation-number undressing gives asymptotic canonical commutators on a fixed polynomial core. The original Hamiltonian has additional directions: its first action has a Poisson(2λ) particle–hole deficit and only a fraction $\sqrt{q}$ of its squared norm lies in the pure-creation direction. This identifies a concrete next calculation, the joint $(H, M, M^\dagger)$ law.

1 Literature-led question and microscopic conventions

The Susskind static-patch program motivates deriving particle operators from the hologram. The quantitative singlet/meson correspondence remains a target in [1, Sec. 4.3]. Here we pursue the state and polarization input following the state-adapted construction of Rajgadia–Xu [2, Sec. 3.2]. Their dressed oscillator representation is a predecessor, not a new general construction claimed in this note. Our task is to derive a microscopic realization with the original complex SYK couplings and then test its Hamiltonian closure.

Use N complex fermions, $\{c_i, c_j^\dagger\} = \delta_{ij}$, and

$$H_N = \sqrt{w_N} \sum_{|A|=|B|=p} J_{AB} C_A^\dagger C_B, \quad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p}\right)^2, \quad \lambda_N = \frac{p^2}{N}. \tag{1}$$

$C_A^\dagger$ is the product of creators in increasing site order and $C_A = (C_A^\dagger)^\dagger$. The centered Hermitian Gaussian matrix has $\mathbb{E} J_{AB} J_{CD} = \delta_{AD}\delta_{BC}$; overlapping A, B are included. All operators below use this same J. Take even N, $m = N/2$, $1 \le p \le m$, $p \to \infty$, $\lambda_N \to \lambda > 0$, fixed $J_s > 0$, and $\hbar = 1$. The Hamiltonian uses string-energy units. All word lengths are fixed before this limit; no long-time or growing-excitation limit is taken.

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2 Keep the reference polarization in the operator

Fix an occupied set O, $|O| = m$, and its empty complement E, $|E| = a = N - m$. Write $\Omega_O = |O\rangle$ and

$$\phi_{O,N}(X) = \mathbb{E}_J \langle O|X(J)|O\rangle. \tag{2}$$

This is an annealed vector state; it is not the full-Fock trace of the preceding bilocal calculations. The reference state is not asserted to be stationary under H_N.

Why changing only the state is insufficient. For a jointly $U(N)$-covariant word $W(J)$, $\mathbb{E}W$ commutes with the $U(N)$ action on $\wedge^m\mathbb{C}^N$. Irreducibility and Schur's lemma give the exact finite-size identity

$$\phi_{O,N}(W) = \mathbb{E}_J \operatorname{tr}_m W, \qquad \operatorname{tr}_m = \binom{N}{m}^{-1} \operatorname{Tr}_{\mathcal{H}_m}. \tag{3}$$

No Haar average of $|O\rangle$ is needed. In particular all annealed H-only moments are unchanged. This explains why the operator must retain information about O.

Define the original Hamiltonian's pure-excitation block and its normalization:

$$V_N^\dagger = \sqrt{w_N} \sum_{\substack{A\subset E,\ B\subset O \\ |A|=|B|=p}} J_{AB} C_A^\dagger C_B, \tag{4}$$

$$L_N = \binom{a}{p}\binom{m}{p}, \qquad \sigma_N^2 = w_N L_N, \qquad M_N^\dagger = V_N^\dagger/\sigma_N, \quad M_N = (M_N^\dagger)^\dagger. \tag{5}$$

The selected J_{AB} are independent circular complex Gaussians of unit variance. Their conjugates occur in V_N. These are a subfamily of the original disorder, not an independent probe ensemble. Every selected monomial preserves the physical total charge Q, while creating p particles in E and p holes in O. Exactly,

$$M_N\Omega_O = 0, \qquad \phi_{O,N}(M_N M_N^\dagger) = 1, \qquad \phi_{O,N}(M_N^\dagger M_N) = 0. \tag{6}$$

The occupation projector is additional reference data. Equation (3) does not apply to a word that keeps this projector fixed while rotating the fermions and couplings.

This choice differs from taking the empty physical complex-fermion vacuum: the number-conserving H_N annihilates that vacuum. In the Majorana KM construction of [2], the analogous spin-flip number is not the conserved charge of (1). The two microscopic constructions therefore require separate pairing derivations.

3 All fixed words: active intervals and backward corrections

Read a word in $M_N, M_N^\dagger$ from right to left. A creation raises the particle–hole block count by one; an annihilation lowers it by one. A balanced word gives a Dyck path if every intermediate height is nonnegative. For a creation at height h, set

$$R_N(h) = \frac{\binom{a-hp}{p}\binom{m-hp}{p}}{L_N}, \qquad q = e^{-4\lambda}, \tag{7}$$

with out-of-range binomials zero.

Theorem 1 (Fixed-word limit)**.** *For each fixed word w, $\phi_{O,N}(w(M_N, M_N^\dagger))$ tends to the vacuum moment of the weighted shift*

$$M|n\rangle = \sqrt{nq^{n-1}}\,|n-1\rangle, \qquad M^\dagger|n\rangle = \sqrt{(n+1)q^n}\,|n+1\rangle \tag{8}$$

on $\ell^2(\mathbb{N}_0)$. For a balanced Dyck word of length $2k$, its finite-size forward-pairing contribution is exactly

$$F_N(w) = \left(\prod_{\text{annihilations}} h\right) \prod_{\text{creations}} R_N(h). \tag{9}$$

The remaining contribution satisfies

$$|\phi_{O,N}(w) - F_N(w)| \le k!\rho_N(k), \quad \rho_N(k) = \frac{\binom{kp}{p}}{\binom{a}{p}}\frac{\binom{kp}{p}}{\binom{m}{p}} \longrightarrow 0. \tag{10}$$

An unbalanced or non-Dyck word has zero vacuum moment already at finite size.

Forward pairings. Complex Gaussian Wick expansion matches each creation coefficient to an annihilation coefficient. A pair is forward if its creation acts first. For an all-forward matching, a label is active between its two endpoints. At each creation, both its particle set and its hole set must avoid every active set. This condition is necessary and sufficient for a nonzero CAR word. Legal active sets are pairwise disjoint; each composite monomial has even degree $2p$, so active monomials commute. Each annihilation removes its own active monomial with positive amplitude. Thus every allowed assignment contributes exactly $+1$.

With h labels active, the next creation has exactly $\binom{a-hp}{p}\binom{m-hp}{p}$ choices. A closing endpoint has h choices of partner. Normalizing every pair by L_N proves (9). Both nested and crossed intervals impose exclusion when simultaneously active. Counting only crossing chords would give the wrong law.

Backward pairings are small, not identically zero

In a matching with a backward pair, choose the first annihilation whose matched creation has not yet acted. Before this endpoint, every nonzero action is an all-forward history. Its particle and hole sets must each lie inside the unions already excited, each of size at most kp. This newly encountered Gaussian label is independent of all previous labels. Its conditional allowed fraction is at most $\rho_N(k)$. All later CAR amplitudes have absolute value at most one. There are at most $k!$ matchings, proving (10). For fixed k and sufficiently large N,

$$\rho_N(k) \le \left(\frac{kp}{a-p+1}\frac{kp}{m-p+1}\right)^p \longrightarrow 0. \tag{11}$$

This argument also allows different Gaussian labels to take the same site subsets; they are not excluded by hand.

For example, the alternating fourth moment has the finite correction

$$\phi_{O,N}(M_N M_N^\dagger M_N M_N^\dagger) = 1 + \frac{1}{L_N}. \tag{12}$$

The extra matching repeats the same particle and hole sets. It is an explicit check against replacing all finite-size pairings by forward ones prematurely.

Taking the limit. For each fixed h,

$$\log R_N(h) = -hp^2\left(\frac{1}{a}+\frac{1}{m}\right) + O_h(p^3/N^2) \longrightarrow -4\lambda h. \tag{13}$$

Consequently (9) tends to

$$F(w) = \left(\prod_{\text{annihilations}} h\right) q^{\sum_{\text{creations}} h} = \prod_{\text{annihilations}} h q^{h-1}. \tag{14}$$

The two height sums agree because each upward edge of a Dyck path has a downward edge at the same level. The shift (8) has exactly these moments. Together with (10), this proves Theorem 1 for all fixed words, not only creation norms.

The creation norms themselves are exact at finite size:

$$s_{k,N} = \phi_{O,N}(M_N^k (M_N^\dagger)^k) = k! \prod_{j=0}^{k-1} R_N(j) \longrightarrow k! q^{k(k-1)/2} =: s_k. \tag{15}$$

There is no backward matching in this ordering. In particular, $\phi(M^2(M^\dagger)^2) = 2q$, whereas $\phi(MM^\dagger MM^\dagger) = 1$. These distinguish the dressed mode from a bare canonical oscillator.

4 Canonical variable and the new vacuum representation

On the standard bosonic finite-particle core, with $b|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{n} = b^\dagger b$, the shift is

$$M = q^{\hat{n}/2} b, \qquad M^\dagger = b^\dagger q^{\hat{n}/2}. \tag{16}$$

The order matters: M acts with $q^{(n-1)/2}$, not $q^{n/2}$. Conversely, on that core,

$$b = q^{-\hat{n}/2} M, \qquad b^\dagger = M^\dagger q^{-\hat{n}/2}, \qquad [b, b^\dagger] = I. \tag{17}$$

The inverse dressing is unbounded. Equation (17) is first a statement about the limiting representation and its dense domain. There is also an actual microscopic realization on fixed excitation levels, as follows.

The bare canonical/dressed distinction already appears in [2, HTML Eqs. (75), (85)–(86)]; the corresponding PDF equations are (3.43), (3.53)–(3.54). Here the single-mode law was obtained from (1)–(4), not from the Majorana mixed-chord rules.

An actual finite-size canonical sequence

Let $Q_E = \sum_{i \in E} c_i^\dagger c_i$, $n_N = Q_E/p$, and $q_N = e^{-4\lambda_N}$. Define finite-size operators

$$b_N = q_N^{-n_N/2} M_N, \qquad b_N^\dagger = M_N^\dagger q_N^{-n_N/2}. \tag{18}$$

They are adjoints, and $[n_N, M_N^\dagger] = M_N^\dagger$ exactly. Every intermediate state obtained from Ω_O along a word has n_N equal to that word's height. This remains true when M_N has components outside the span of pure-creation vectors. For a Dyck word define $A(w) = \sum_{\text{annihilation}} (h-1)$. Each creation contributes $q_N^{-h/2}$ and each annihilation $q_N^{-(h-1)/2}$, so, sample by sample,

$$\langle O | w(b_N, b_N^\dagger) | O \rangle = q_N^{-A(w)} \langle O | w(M_N, M_N^\dagger) | O \rangle. \tag{19}$$

Theorem 1 then yields all fixed canonical vacuum moments:

$$\lim_N \phi_{O,N}(w(b_N, b_N^\dagger)) = \prod_{\text{annihilation}} h. \tag{20}$$

Non-Dyck or unbalanced words remain zero. Expanding a squared norm in finitely many such words proves, for every fixed noncommutative polynomial P,

$$\mathbb{E}_J \left\| ([b_N, b_N^\dagger] - I) P(b_N, b_N^\dagger) \Omega_O \right\|^2 \longrightarrow 0. \tag{21}$$

Thus the canonical sequence is microscopic, with convergence on the specified Slater polynomial core. The norm of the inverse dressing on the full Fock space grows with N; no uniform operator-norm limit or exact finite-matrix CCR is asserted.

Actual ladder vectors and the limiting algebra

For fixed k and sufficiently large N such that $kp \le \min(a, m)$, normalize $|k; N\rangle = (M_N^\dagger)^k \Omega_O / \sqrt{s_{k,N}}$ in the annealed state/disorder Hilbert space. Different k are orthogonal by their particle–hole excitation numbers. Creation maps one to the next exactly, with coefficient $\sqrt{s_{k+1,N}/s_{k,N}}$. The all-word theorem gives

$$\left\| M_N |k; N\rangle - \sqrt{s_{k,N}/s_{k-1,N}}\, |k-1; N\rangle \right\|^2 \longrightarrow 0 \qquad (k \ge 1). \tag{22}$$

Indeed, the overlap with $|k-1; N\rangle$ equals $\sqrt{s_{k,N}/s_{k-1,N}}$ exactly, and the norm squared of the first term tends to kq^{k-1} by a fixed word of length $2k+2$. Their difference therefore vanishes. This is fixed-level annealed norm convergence, not operator-norm convergence on the entire microscopic Hilbert space.

Algebra generated by the adapted mode. The weights in (8) are positive and tend to zero; M is a compact bounded operator. Its kernel is exactly $\mathbb{C}|0\rangle$, so the vacuum projection $P_0 = 1_{\{0\}}(M^\dagger M)$ belongs to $W^*(M)$. The operators $(M^\dagger)^n P_0 M^j$ are nonzero multiples of every matrix unit $|n\rangle\langle j|$. Thus

$$W^*(M) = B(\ell^2(\mathbb{N}_0)). \tag{23}$$

The vacuum is cyclic and not separating. This is a new adapted algebra; it does not split the old tracial field algebra into physical annihilators. The separating tracial chord vacuum of [3] refers to its specified field algebra, not all auxiliary Fock ladders.

The difference of states is already visible at finite size. Averaging the Gaussian coefficients and taking the normalized full-Fock trace gives

$$\mathbb{E}_J \operatorname{tr}_{\text{full}}(M_N^\dagger M_N) = 4^{-p}, \qquad \phi_{O,N}(M_N M_N^\dagger) = 1. \tag{24}$$

Each diagonal term imposes p occupied and p empty sites, so its normalized trace is 2^{-2p}. The same microscopic sequence therefore vanishes in the old tracial L^2 norm and survives in the adapted state. It is not an observable identified by a state-preserving map from the old tracial limiting algebra.

Fixed scalar words also concentrate. This is not needed for the annealed theorem, but follows from the same label argument. For $F_{w,N}(J) = \langle O | w(M_N, M_N^\dagger) | O \rangle$ of length $2k$, expand the variance in two vacuum copies. Wick matchings internal to each copy cancel. In every remaining matching a shared label appears only once in each copy. All of its sites must occur in the other labels, or that copy cannot return to the vacuum. Conditioning on at most $2k-1$ other labels gives

$$\operatorname{Var} F_{w,N} \le (2k)! \rho_N(2k-1) \longrightarrow 0. \tag{25}$$

For the microscopic b_N words, (19) multiplies this bound by a fixed-word factor tending to $q^{-2A(w)}$. No concentration claim for the joint Hamiltonian dynamics is needed or inferred here.

5 The full Hamiltonian: a Poisson family of missing directions

The adapted mode captures an actual part of $H_N\Omega_O$. To measure the rest, let k be the number of particles placed in E by a term of H_N (and thus the number of holes in O). The exact annealed squared norm in this sector is

$$W_{k,N} = w_N \binom{m}{p}\binom{p}{k}\binom{a}{k}, \qquad 0 \le k \le p. \tag{26}$$

To see this, B must be an occupied p-set. After removing it, the creator set A can fill $p-k$ of its p holes and k of the a empty sites. Squaring and averaging pairs a term only with its Hermitian reverse; no independent-probe approximation is made. Vandermonde's identity gives

$$C_{m,N} = \phi_{O,N}(H_N^2) = w_N \binom{m}{p}\binom{a+p}{p} \longrightarrow C = J_s^2/\lambda. \tag{27}$$

For the deficit $r = p-k$, the normalized weights and their falling factorial moments are

$$\Pr_N(r) = \frac{\binom{p}{r}\binom{a}{p-r}}{\binom{a+p}{p}}, \tag{28}$$

$$\mathbb{E}_N(r)_{\underline{t}} = \frac{(p)_{\underline{t}}^2}{(a+p)_{\underline{t}}} \longrightarrow (2\lambda)^t. \tag{29}$$

This is a hypergeometric law: choose p items from $a+p$, of which p are marked. For $t \le p$, its factorial moments are bounded by $(p^2/a)^t$. Hence for every fixed $z \ge 1$, $\mathbb{E}_N z^r \le \exp[(z-1)p^2/a]$. The factorial-moment expansion is uniformly dominated, identifying the limit as Poisson(2λ). In particular,

$$\frac{\sigma_N^2}{C_{m,N}} = \frac{\binom{a}{p}}{\binom{a+p}{p}} \longrightarrow e^{-2\lambda} = \sqrt{q}. \tag{30}$$

This is a squared-norm fraction. The normalized overlap amplitude between $H_N\Omega_O$ and $M_N^\dagger\Omega_O$ tends to $e^{-\lambda} = q^{1/4}$.

Let $\mathcal{K}_N$ be the closed span of $(M_N^\dagger)^j\Omega_O$ in the annealed joint Hilbert space. At finite N, $H_N\Omega_O$ overlaps only its one-block vector: the mean vacuum overlap vanishes and higher vectors have more than p excited particles. Therefore exactly

$$P_{\mathcal{K}_N} H_N\Omega_O = \sigma_N M_N^\dagger\Omega_O, \tag{31}$$

$$\|(1-P_{\mathcal{K}_N})H_N\Omega_O\|^2 = C_{m,N} - \sigma_N^2 \longrightarrow C(1-\sqrt{q}) > 0. \tag{32}$$

The physical H_N does not close on this single ladder. A guessed number Hamiltonian would lose the nonzero norm in (32).

6 Exact checks and the next constructive calculation

The new driver `code/check_slater_ladder.py` uses integer CAR actions and rational Gaussian Wick sums. Its first successful output is preserved in the stage's `exact-checks.json`; it was not rerun to overwrite an earlier result.

Finite fixtures $(N, p; k_{\max})$	$(4,1;4),(6,1;3),(6,2;3),(8,2;3)$
Gaussian label assignments	1,537,310
Forward CAR/active-interval comparisons	754,030
Complete Dyck-word moments	46
Non-Dyck words vanishing by right action	136
Words with nonzero backward corrections	25
Exact hypergeometric diagnostic rows	6
Mixed-Q linear-cumulant checks	189

The fixtures also enumerate every original p-body monomial in $H_N\Omega_O$, independently checking (26) and (27). The hypergeometric rows use $\lambda_N = 1/2$, $p = 2, 4, 8, 16, 32, 64$, $N = 2p^2$. Saved total-variation distances to Poisson(1) are floating-point diagnostics; the exact factorial moments and the analytic bound above supply the convergence argument. Small fixtures test identities and signs; they do not prove the double-scaling limit.

The next microscopic object is now specified: preserve the same J and compute the joint ordered moments of $H_N, M_N, M_N^\dagger$, including the $r \geq 1$ sectors of (28). The first target is a positive representation that reproduces (31)–(32) and the interleaved fourth moments. A subsequent step must connect this adapted algebra to the already constructed singlet bilocal observables. The chosen Slater state is not stationary, so its preparation and real-time evolution are part of that calculation.

Only after these steps can one test a space/boost generator and a meson mass operator against the proposed bulk theory. The reference polarization needed for canonical bilocals in light-front QCD is explicit in Itakura and Rajeev [4, 5]; it supplies a useful comparison, not an identification of our block number with their momentum or mass variables. Neither a finite-G interaction nor a complete de Sitter dictionary has been derived here.

Reproducibility. The JSON records parameters, versions, parameter digest, driver hash and parent-manifest hashes. The local evidence recorder additionally binds sources, proofs, manuscript, two compiler passes and all final page renders. It verifies preserved files and saved arithmetic; it does not certify a theorem. The prior bilocal scientific outputs, including historical failures in the separate KS branch, are retained.

A A supporting check on the old tracial low fields

The preceding three-species law has normalized self-adjoint fields (h, u, x) with covariance I and

$$Q = \begin{pmatrix} q & q & q^2 \\ q & q & q^2 \\ q^2 & q^2 & q^4 \end{pmatrix}. \tag{33}$$

For real $Y = \alpha h + \beta u + \gamma x$, set $A = \alpha^2 + \beta^2$, $B = \gamma^2$. Its fourth classical cumulant is the direct mixed-Q Wick consequence

$$\tau(Y^4) - 3\tau(Y^2)^2 = -(1-q)A^2 - 2(1-q^2)AB - (1-q^4)B^2. \tag{34}$$

It is strictly negative for every nonzero Y at fixed $0 < q < 1$, and lies between $(q^4-1)(A+B)^2$ and $(q-1)(A+B)^2$. Thus no nonzero linear whitening of this triple is a Gaussian field in a quasifree canonical state. This is an elementary implication of the previously obtained law, not a new mixed-Q theorem and not an obstruction to every nonlinear or nonquasifree representation. It motivates retaining explicit polarization rather than only changing a low-field Gram matrix.

The companion KMS note separately treats joint annealed Gibbs tilting, exact thermofield realization, and smooth frequency filters. Its Gaussian dilation matches two-point functions only. None of these constructions is substituted for the same-J Slater derivation in the main text.

References

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